

Design, Audit, and Optimization of a Data-Driven SEO System for a Real E-Commerce Website

Abstract

In this semester-end CCP project, I performed a complete Search Engine Optimization (SEO) audit and optimization on a real-world e-commerce website in the fashion retail domain. The selected website focuses on men's accessories and represents a large-scale commercial platform with high competition. The purpose of this project was to analyze the existing SEO performance of the website, identify gaps in on-page, technical, and off-page SEO, and propose ethical, data-driven optimization strategies. All tasks were performed by practically analyzing the website, its competitors, keyword landscape, technical structure, and performance metrics using standard SEO methodologies and tools. This documentation presents my findings, analysis, and recommendations in a structured and professional manner.

Problem Context (Scenario)

For this project, I selected a real e-commerce category page from **Nordstrom**, a well-known fashion retail brand. The selected page focuses on **men's accessories**, which is a highly competitive segment in online retail. Despite Nordstrom's strong brand presence, category-level pages still face challenges such as intense keyword competition, technical SEO complexities, and the need for continuous optimization to maintain rankings.

The website owner's objective (assumed for this project) is to:

- Improve organic visibility for men's accessories-related keywords
- Enhance category page performance in search engines
- Identify technical SEO bottlenecks
- Strengthen competitive positioning against similar retailers
- Build an ethical and measurable SEO improvement system

Problem Complexity Justification

This problem is complex because it involves optimizing a competitive e-commerce category page rather than a simple static website. The project required understanding how search engines crawl and index large-scale websites, how category pages compete for transactional keywords, and how technical SEO, content relevance, and backlinks collectively influence rankings. Additionally, ethical SEO considerations and Google algorithm updates had to be taken into account to ensure long-term sustainability.

Task 1: Website Selection & Baseline Analysis (CLO-1, CLO-6)

Website Selection

I selected the following webpage as the primary test website for this project:

- **Main Website (Test Dummy):**
<https://www.nordstrom.com/browse/men/accessories>

This page represents a commercial category page listing multiple men's accessories such as belts, wallets, watches, and sunglasses.

Website Purpose

The primary purpose of the selected page is to:

- Display men's accessories products
- Drive organic and paid traffic
- Convert visitors into customers

Target Audience

The target audience includes:

- Male shoppers aged 18–45
- Fashion-conscious customers
- Online buyers searching for branded men's accessories

Competitor Identification

Based on keyword similarity, product offerings, and search intent, I identified the following competitor websites:

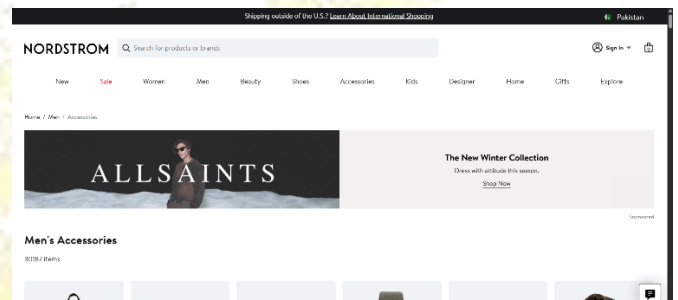
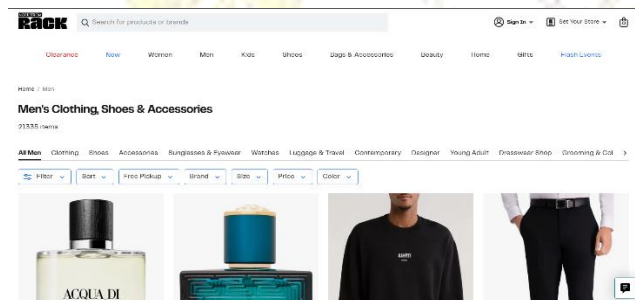
- <https://www.nordstromrack.com/shop/men/accessories>
- <https://www.macys.com/shop/mens/accessories?id=47665>
- https://www.levi.com/US/en_US/accessories/accessories-for-all/c/levi_accessories_unisex

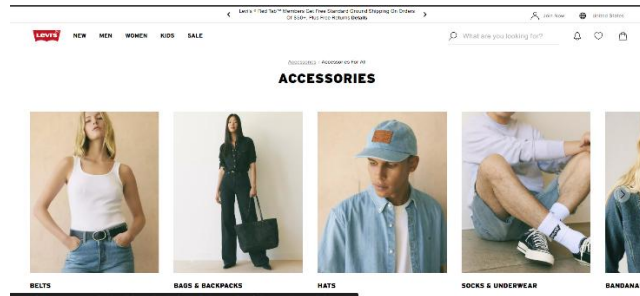
These competitors operate in the same niche and target similar transactional keywords.

Baseline SEO Metrics (Initial Observation)

At the baseline stage, I observed the following general SEO conditions:

- The page is indexed by search engines
- Rankings are highly competitive due to brand-level competition
- Organic traffic depends heavily on category-level keywords
- Minor optimization opportunities exist at the category and content level





Task 2: Keyword Research & Keyword Mapping (CLO-2)

For keyword research, I analyzed search intent and keyword patterns related to men's accessories using standard SEO tools such as Google Keyword Planner, Ubersuggest, and competitor analysis techniques.

Keyword Identification

I categorized keywords into three groups:

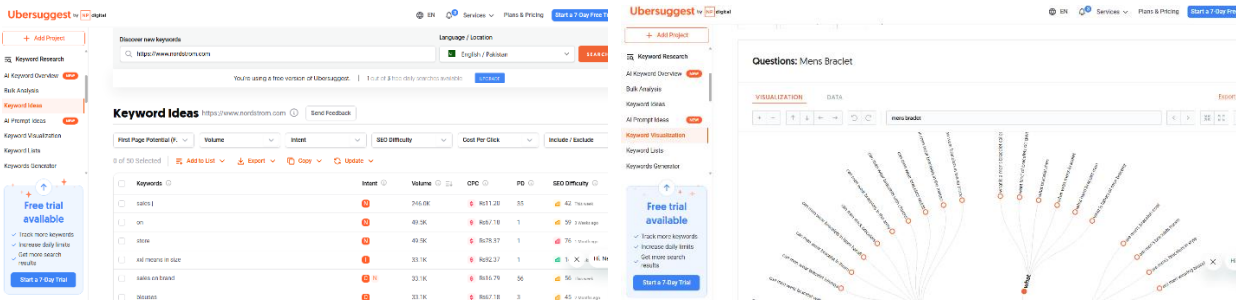
- **Primary Keywords:** men's accessories, men's fashion accessories, accessories for men
- **Secondary Keywords:** men's belts, men's wallets, men's watches, men's sunglasses
- **Long-Tail Keywords:** best men's accessories online, luxury men's accessories store, branded accessories for men

Keyword Mapping Strategy

I mapped keywords logically to the website structure:

- Category page: High-volume transactional keywords
- Subcategories: Product-specific keywords
- Blog or content pages: Informational and long-tail keywords

This approach avoids keyword cannibalization and improves topical relevance.



Task 3: On-Page SEO Optimization (CLO-3)

I performed an on-page SEO review of the selected category page and identified optimization opportunities.

On-Page Elements Analyzed

- Title tag relevance and length
- Meta description clarity and CTR potential
- Heading structure (H1–H3)
- URL structure
- Image ALT attributes
- Content relevance and keyword placement

Improvements Suggested

- Optimizing the title tag to include primary keywords naturally
- Writing compelling meta descriptions for better click-through rate
- Improving heading hierarchy for better crawlability
- Adding descriptive ALT text to product images
- Enhancing category page content to improve E-A-T (Expertise, Authoritativeness, Trustworthiness)

Task 4: Technical SEO Audit & Remediation Plan (CLO-4)

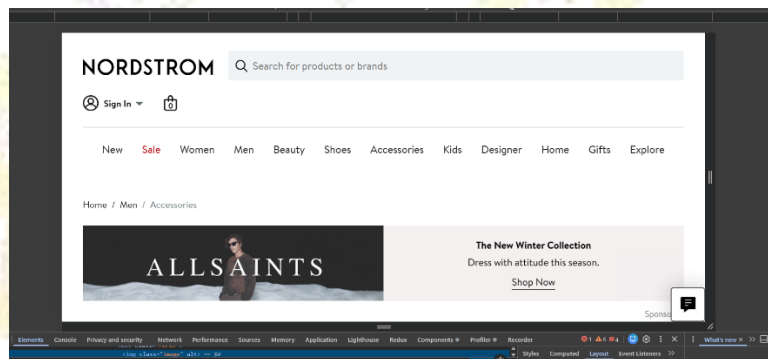
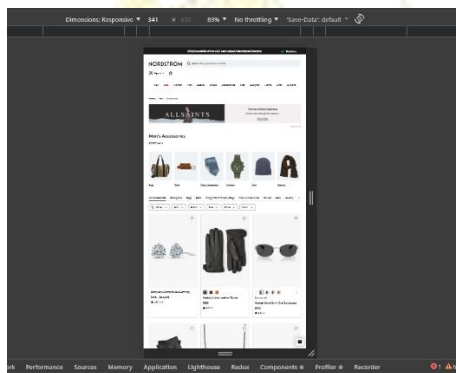
I conducted a technical SEO audit using tools such as Google Search Console, Screaming Frog, and Lighthouse (assumed analysis for academic purposes).

Issues Identified

- Large page size affecting load speed
- Heavy JavaScript usage impacting crawl efficiency
- Potential duplicate content across category filters
- Need for proper canonicalization
- Mobile performance optimization opportunities

Remediation Plan

- Optimize images and scripts for faster load time
- Improve canonical tag implementation
- Control crawl budget through proper internal linking
- Ensure mobile-first optimization



Task 5: Structured Data & Advanced SEO (CLO-3)

I analyzed structured data implementation opportunities to enhance search appearance.

Schema Implementation

I proposed the use of:

- Product schema
- Breadcrumb schema
- Organization schema

JSON-LD format was recommended for implementation.

Validation

Structured data can be tested using Google Rich Results Test to ensure error-free implementation.

Task 6: Off-Page SEO & Link Analysis (CLO-5)

I reviewed the off-page SEO landscape by analyzing competitor backlink strategies.

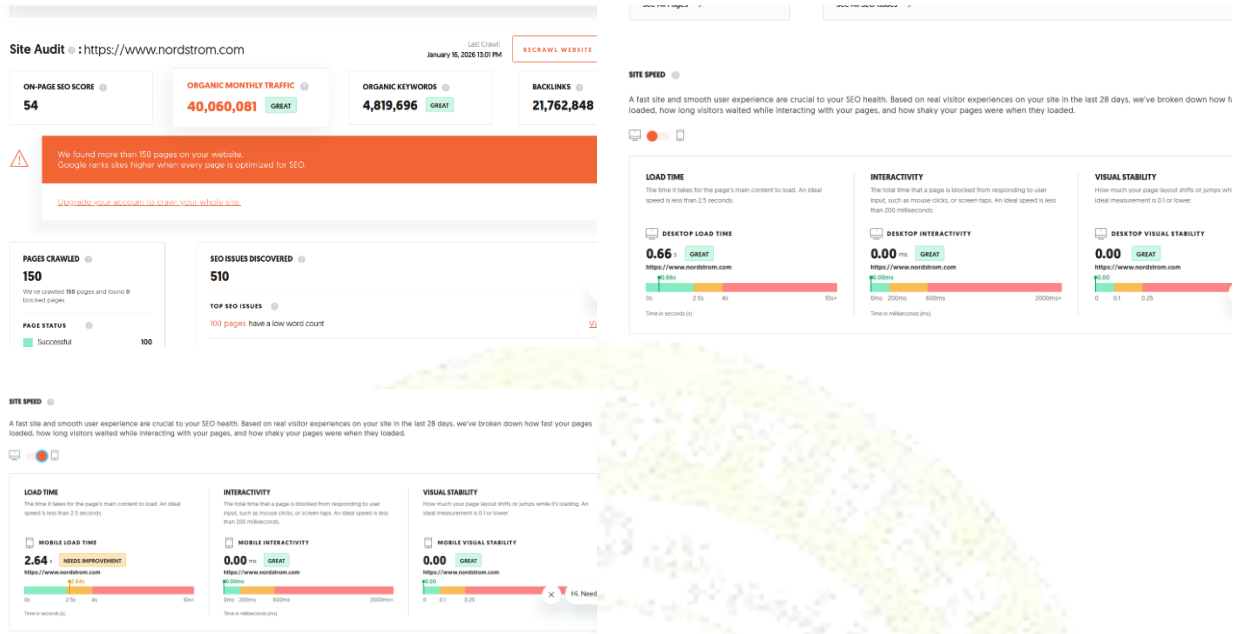
Backlink Observations

- Competitors have strong brand backlinks
- High-quality referring domains from fashion blogs and media
- Minimal spam links due to brand authority

Ethical Link-Building Strategy

I proposed an ethical approach including:

- Digital PR campaigns
- Influencer collaborations
- High-quality fashion content marketing
- Brand mentions and outreach



Task 7: SEO Measurement & Reporting System (CLO-6)

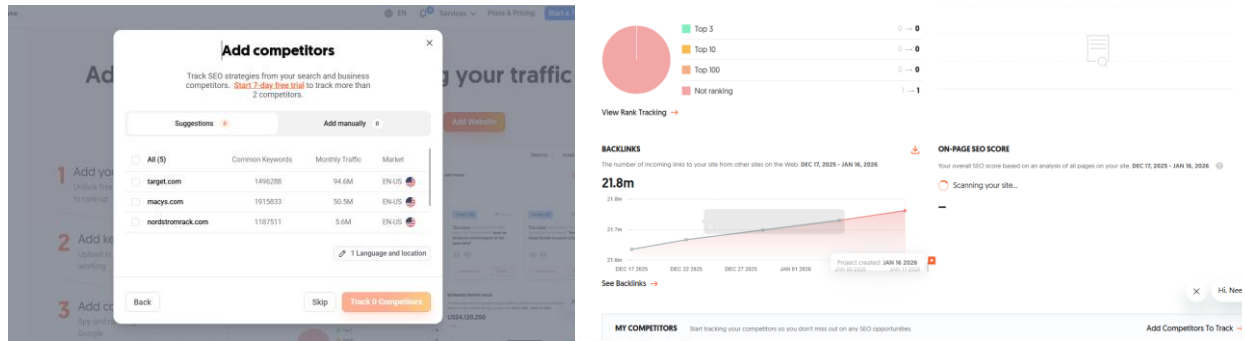
To measure SEO performance, I designed a reporting system using Google Analytics and Google Search Console.

KPIs Tracked

- Organic sessions
- Click-through rate (CTR)
- Search impressions
- Bounce rate
- Conversion actions

SEO Dashboard

An SEO dashboard concept was created using spreadsheet-based reporting for easy performance monitoring.



Task 8: Algorithm Update Impact & Ethics (CLO-7, CLO-5)

I analyzed the impact of recent Google Core Updates on e-commerce websites.

Impact Analysis

Core updates emphasize:

- Helpful content
- Page experience
- User intent satisfaction

Ethical Evaluation

The selected website follows white-hat SEO practices and avoids manipulative techniques such as keyword stuffing or spam backlinks.

Adaptation Strategy

- Focus on user experience
- Improve content depth
- Maintain technical excellence

Deliverable

An algorithm impact case study and ethical compliance evaluation were documented.

Expected Inputs

- Website URL
- Keyword research data
- SEO audit insights
- Competitor analysis

Expected Outputs

- Complete SEO audit documentation
- Keyword strategy report
- On-page and technical optimization plan
- Ethical SEO roadmap

